

Chapter 6

THE ABDOMEN

Segmentation of the Abdomen

The abdomen (Fig. 1) is composed of a series of segments which are more equally developed than in the other regions of the body (Matsuda, 1975). For the most part they retain their simple annular form, the terga and sterna are generally undivided shields, while the pleura are membranous and usually without differentiated sclerites. Each intersegmental sclerite is believed to have fused with the segmental plate behind it. Reduction or

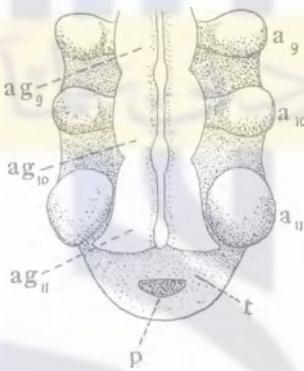


FIG. 41
Ventral view of last abdominal segments of young embryo of *Gryllotalpa*. From Heymons

a_9 to a_{11} , appendages of 9th to 11th segments and ag_9 – ag_{11} , neuromeres of those segments; p , proctodaeum; t , telson.

special modification of certain segments is evident at the anterior and posterior ends of the abdomen, more especially in the latter region, and this specialization increases from the lower to the higher orders. The primitive number of abdominal segments as revealed by embryology is eleven, with a terminal non-segmental region or telson (Anderson, 1972). The telson is present in the embryos of certain insects (Fig. 41), but it rarely persists as a discrete region; it is evident, however, in the Protura, while traces are found in other insects. The 11th segment is present in the adults of the lower orders where its tergum is represented by the *epiproct* above the anus (often fused with the 10th tergum), while vestiges of its sternum are seen in the *paraprocts* which lie on either side of the anus (Fig. 42). The 10th segment is usually distinct and forms the terminal segment in the higher orders. The Protura differ from all other hemimetabolous insects in that the number of abdominal segments increases during post-embryonic development, the youngest instars having only eight segments and a telson. The Collembola

are also exceptional in never having more than six abdominal segments, either in the embryo or the adult. In most insects the 1st abdominal segment, and more especially its sternum, is reduced or vestigial. Among Apocritan Hymenoptera this segment becomes fused with the metathorax during the change from the larva to the pupa and is known as the *propodeum*, *epinotum* or *median segment*.

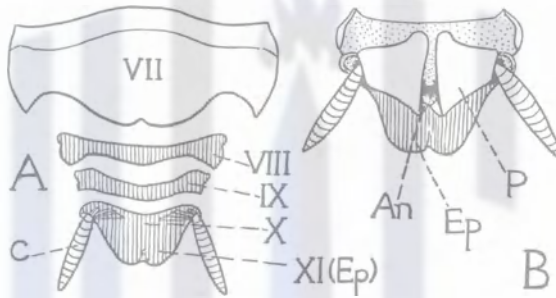


FIG. 42 Last abdominal segments of *Blatta*. A. Dorsal view. B. Ventral view

An, anus; C, cercus; Ep, epiproct; P, paraproct; VII–XI, terga. After Snodgrass.

In many Endopterygote insects, more especially those whose eggs are deposited within plant tissues or in other concealed situations, the distal abdominal segments become attenuated and often telescoped to form a retractile tube or *ovisclapt* which is used as an ovipositor. This modification is particularly well exhibited in the Cerambycidae, Cecidomyidae, Trypetidae, Muscidae and other families. A true ovipositor is usually regarded as appendicular and is dealt with in the next section.

Appendages and Processes of the Abdomen

In the embryos of most insects evident rudiments of paired abdominal appendages appear at some stage during development. They are commonly present on each of the eleven segments in the less specialized Exopterygotes but do not occur on the telson (Fig. 41). A variable number of these appendages may become transformed into organs that are functional during post-embryonic life while the remainder disappear. The most conspicuous of the persistent appendages are the *cerci* of the 11th segment, which exhibit wide diversity of form and may even be transformed into forceps, as in the Japygidae and the earwigs. Among the Apterygota the retention of at least some abdominal appendages is a general feature. They are well exhibited in the Machilidae, where they are present in a reduced condition on the 2nd to 9th segments, and as cerci on the 11th segment. The reduced appendages each consist of a limb-base or *coxite* bearing a distal *stylus* which is

sometimes regarded as the vestige of the shaft of a typical walking limb but which may conceivably be a 'coxal epipodite' homologous with the coxal styles of Symphytan or Machilid legs. In the Diplura the coxites fuse with the sterna and the styli arise directly from the composite plates so formed (Fig. 43). In most insects, however, the styli disappear so that the typical abdominal sternum of insects is probably a plate of compound origin — a 'coxosternum'. In adult Pterygota the cerci are retained in the Palaeopteran and Orthopteroid orders but pregenital abdominal appendages are absent and the only other abdominal appendages present are the *gonopods* which are

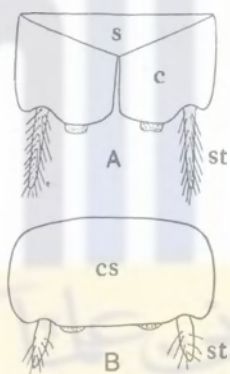


FIG. 43

Abdominal sterna and appendages of A, *Machilis* and B, *Campodea*

s, sternum; c, coxite; cs, coxosternum; st, stylus.

believed by many to enter into the formation of the *genitalia* or external reproductive organs. These are associated with the 8th and 9th segments in the female and with the 9th segment in the male. A completely developed gonopod consists of a coxite bearing a distal stylus, while on its medial border the coxite is produced into a tubular outgrowth or *gonapophysis* (Fig. 44). The term 'gonapophysis' is sometimes also used loosely to denote any genital appendage. The genital aperture is usually located in the membrane immediately behind the 8th or 9th segment.

Among immature Pterygote insects, the gills of mayfly nymphs, the abdominal prolegs of Lepidopteran and Symphytan larvae, the terminal appendages of Trichopteran larvae and the gills of Sialoid larvae have all been regarded as true abdominal appendages (Snodgrass, 1935), but this is not satisfactorily established (Pryor, 1951). On the one hand, the prolegs or gills develop from what seem to be the abdominal appendages of the polypod embryo (Friedman, 1934). On the other hand in the Lepidoptera and other Panorpoïd orders the prolegs appear to have arisen independently in each group, they are absent or lack muscles in the more primitive Panorpoïds, and only in the Lepidoptera do they become important locomotor structures (Hinton, 1955). Examples of what are certainly secondary abdominal processes include the gills of the Trichoptera and most Plecoptera, the pseudopods and creeping welts found in some Dipteran larvae (Hinton, 1955) and the copulatory organs on the 2nd abdominal segment of male Odonata.

External Genitalia

These include the characteristic ovipositor of many female insects and the diverse male copulatory organs. The morphological interpretation of these structures presents many problems, especially as there is a basic and long-standing controversy as to whether they are, at least in part, modified abdominal appendages serially homologous with the thoracic limbs, or whether they are simply secondary sternal processes. Many taxonomists have preferred to avoid a terminology based on uncertain homologies, so that a large number of special anatomical terms are now applied to the external genitalia of insects (Tuxen, 1970). In the above brief description of a gonopod we have, in effect, interpreted the genitalia as appendicular derivatives, with homologies existing between the various components of the male and

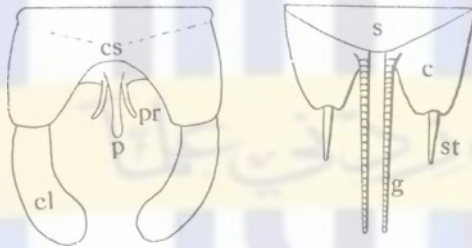


FIG. 44
Left: Diagram of typical male genitalia. Right: A pair of primitive gonopods

c, coxite; *cl*, clasper; *cs*, coxosternum; *g*, gonapophysis; *p*, penis; *pr*, paramere; *s*, sternum; *st*, stylus.

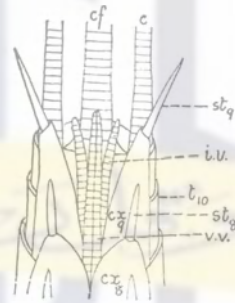


FIG. 45
Ventral view of the apex of the abdomen of a female *Machilis* showing genitalia *c*, cercus; *cf*, median caudal filament; *cx₈*, *cx₉*, coxites of 8th and 9th sterna; *st₈*, *st₉*, styli; *i.v.*, *v.v.*, inner and ventral valves of ovipositor; *t₁₀*, 10th tergum. After Walker, *Ann. ent. Soc. Amer.* 15.

female organs. The matter is by no means settled, however, and further information and discussion can be found in recent summaries of the various theories by Michener (1944), Gustafson (1950), Nielsen (1957), Snodgrass (1957), Matsuda (1958), Gouin (1962), Scudder (1961, 1971) and Smith (1969). In general, the suggested homologies usually start from the structures found in the Thysanuran genitalia and depend also on evidence from postembryonic development, the structure of gynandromorphs, and the anatomical relations, musculature and innervation of the various parts. Scudder (1964), however, has emphasized the need to apply modern concepts of homology based on experimental studies of development; such an approach, combined perhaps with the use of genetic markers for homologous segmental regions may eventually help to resolve some of the persistent difficulties in this field (Sokoloff and Hoy, 1968; Fletcher, 1970).

The Female Genitalia consist typically of three pairs of valves which

collectively form the *ovipositor* or egg-laying organ. Their degree of development and co-adaptation varies according to the uses to which that organ is subjected. In the Mallophaga and Siphunculata, for example, the ovipositor is vestigial or absent; in the Dictyoptera its valves are small and free; in the Tettigoniidae (Figs. 46 and 47) those of one side are held together by tongue and groove joints and form, along with their counterparts of the opposite side, an elongate and powerful egg-laying instrument; in most Hymenoptera (Fig. 47) the ovipositor is greatly attenuated and modified for piercing or stinging. A complete Pterygote ovipositor, according to the interpretation of Scudder (1971), consists of (1) a pair of small coxites of the 8th abdominal segment, each bearing (2) a gonapophysis, which forms an anterior (ventral

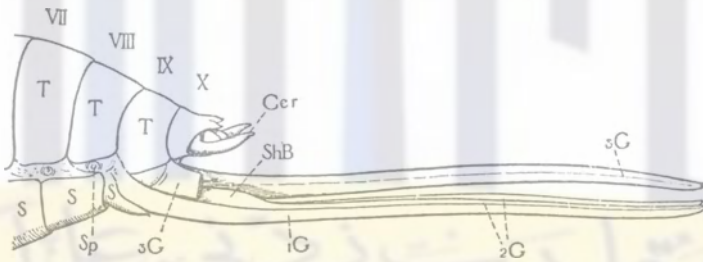


FIG. 46 Ovipositor of a long-horned grasshopper (*Conocephalus*)

VII–X, terga; s, sterna; Cer, cerci; 1G, ventral valve; 2G, inner valves; ShB, bulb-like swelling formed by the union of the bases of 2G; 3G, dorsal valves, the left one is shown as if cut off near its base. After Snodgrass, *U.S. Bur. Ent. Tech. Ser.* 18.

or first) ovipositor valve; (3) a pair of small coxites of the 9th segment, which bear (4) the paired gonapophyses that form the inner (posterior or second) ovipositor valves, as well as (5) a coxal extension or *gonoplac*, forming the dorsal (lateral or third) ovipositor valve. In addition there is on each side a characteristic sclerite (6) the *gonangulum* which is derived from the anterior part of the coxite of the 9th abdominal segment and articulates primitively with this coxite, the 9th abdominal tergum and the gonapophysis of the 8th segment. Essentially the same structures are found among the Thysanura in the Lepismatidae, though the coxites are unmodified there and bear styles. In the Machilidae the gonangulum has not evolved, so that the appendages of the genital segments are rather similar to those of the pregenital region of the abdomen, with the gonapophyses corresponding in position to the eversible vesicles (see p. 435).

The **Male Genitalia** present a wide range of variation and are particularly valuable for separating the genera and species of many groups of insects. When fully developed (Figs. 44 and 48) they consist essentially of a pair of *claspers* which help to grip the female during copulation and between which lies the *aedeagus*. The latter is composed of a *penis* which is usually unpaired (but may be double, as in the Ephemeroptera and some

Dermaptera) and a pair of more lateral structures which are perhaps best referred to as *parameres*, though this term has been used in several different senses. The homologies of these parts have been the subject of much controversy. The claspers probably represent the coxites and styles of the 9th segment, though in some cases they seem to be composed only of the styles, the coxites having fused with the sternum. Snodgrass (1957) has argued that the claspers of the Hemiptera and the Endopterygote insects are non-

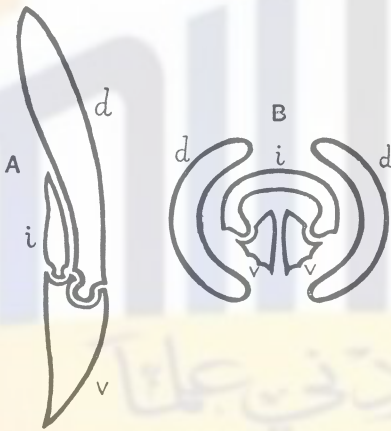


FIG. 47
Transverse sections of the ovipositor of: A, an Orthopteran (*Tettigonia*) after Dewitz; B, a Hymenopteran (*Sirex*) after Taschenberg. The method of interlocking of the valves is shown
d, dorsal valve; i, inner valve; v, ventral valve.

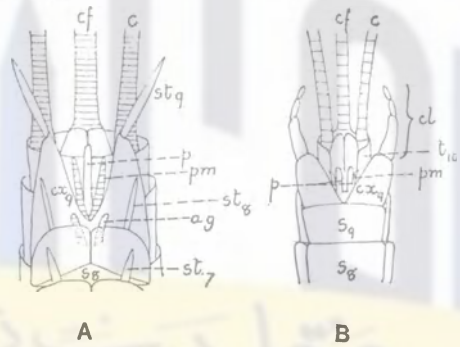


FIG. 48
Ventral view of the apex of the abdomen of A, *Machilis* and B, an Ephemerid showing male genitalia
ag, gonapophyses of 8th segment; c, cercus; cf, median caudal filament; cl, clasper; cx₉, coxite of 9th sternum; p, penis (paired in B); pm, paramere; s₈ s₉, 8th and 9th sterna; st₇-st₉, 7th to 9th styli; t₁₀, 10th tergum. After Walker, *Ann. ent. soc. Am.* 15.

appendicular parameres, formed from the paired phallic rudiments of the 10th abdominal segment. Part or all of the aedeagus is held by some authors to be of a secondary, non-appendicular nature but others consider it to be formed by the division of the gonapophyses of the 9th abdominal segment, the two median halves fusing during development to form the penis while the lateral halves constitute the parameres. Yet another view regards it as formed in a similar manner from the gonopods of the 10th abdominal segment.

Deviations from the typical structure of the genitalia of both sexes are described in the sections dealing with the anatomy of the various orders – they are sometimes supplemented by secondary formations or show various degrees of reduction. In addition to the general works cited above, recent studies of the morphology and development of the genitalia of particular groups of insects include the following: Qadri (1940), Snodgrass (1941), Neumann (1958), Dupuis (1955), Klier (1956), Scudder (1959), Davies (1961), Menees (1963), Salzer (1968), Smith, E. L. (1970) and Häfner (1971).